Fiscal Year:	FY 2013	**Task Last Updated:**	FY 07/03/2013
PI Name:	Derby, Jeffrey J. Ph.D.		
Project Title:	Modeling of Particle Transport in the Melt and Its Interaction with the Solid-Liquid Interface		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Minneapolis	**State:**	MN
Zip Code:	55455-0132	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	09/01/2010	**End Date:**	08/31/2015
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
Contact Email:	patton.downey@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date is 8/31/2015, per NSSC information (Ed., 6/3/14) NOTE: End date is 8/31/2014, per NSSC information (Ed., 5/30/2013)		
Key Personnel Changes/Previous PI:	Jeffrey Derby is the PI of this NASA project. Collaborators funded by the DLR ParSiWal project are Jochen Friedrich (Fraunhofer IISB), Arne Croell (University of Freiburg), and Heike Emmerich (University of Bayreuth).		
COI Name (Institution):			
Grant/Contract No.:	NNX10AR70G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The primary goal of this research project is to understand the mechanisms that determine the dynamics of particle transport in silicon melts and the pushing or engulfment of these particles at a moving solid-liquid interface, specifically small silicon carbide and silicon nitride particles during the solidification of silicon. This understanding is vital to improving the quality of solar-grade, multi-crystalline silicon grown by directional solidification processes. This work will coordinate with the ParSiWal (Partikeleinfang bei der Siliziumkristallisation im Weltall) project funded by the German DLR (Deutsches Zentrum für Luft- und Raumfahrt) and involving Jochen Friedrich (Fraunhofer IISB), Arne Croell (University of Freiburg), and Heike Emmerich (University of Bayreuth). Specifically, finite-element, moving-boundary analyses are being developed and applied to assess the dynamics of particle pushing and engulfment. Of particular interest is to ascertain critical velocities for engulfment for a variety of different particle sizes and thermal gradients. We will also study the effects of shear and solidification flows, in the melt near the interface, on engulfment mechanisms and the effect of a grain boundary at the interface that locally modifies its shape and energy. Planned experiments by ParSiWal in the laboratory as well as during the TEXUS 50 mission are aimed to probe the effects of particle engulfment during silicon solidification and will be used to assess theoretical predictions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the dynamics of particle pushing and engulfment will enable process development to minimize their effects on grown material. This will promote lower cost and higher quality of solar-grade, multi-crystalline silicon and will advance progress in commercial solar cell production.
Task Progress:	The graduate student supported by this project, Mr. Yutao Tao, has done well in his required coursework during the academic year and recently passed the entrance exam for the doctoral program in Materials Science and Engineering. Model development is ongoing, with changes to our code being made to allow for computation of van der Waals forces, pre-melting phenomena, and the Gibbs-Thomson effect of solidification interfacial curvature. We are focusing on model verification and validation and will soon be conducting a variety of parametric studies of particle engulfment phenomena. Our goal for the coming summer is to write a first paper describing our modeling approach and presenting initial results. Graduate student Tao will present a poster of his research at both the 15th International Summer School on Crystal Growth (ISSCG-15) and the 17th International Conference on Crystal Growth and Epitaxy (ICCGE-17) in August 2013 (see details below). PI Derby will present this poster at the 19th American Conference on Crystal Growth in July 2013 (see below). Derby will also attend the ISSCG-15 and ICCGE-17 meetings and will meet with German ParSiWal participants, Jochen Friedrich (Fraunhofer IISB) and Heike Emmerich (University of Bayreuth), after the conference to discuss initial results and continuing collaboration on this project. Conference presentations in report period: Y. Tao, A. Yeckel , and J.J. Derby, “Modeling of Particle Engulfment during the Growth of Multicrystalline Silicon for Solar Cells,” Poster presentation, 19th American Conference on Crystal Growth and Epitaxy (ACCGE-19) and 16th U.S. Biennial Workshop on Organometallic Vapor Phase Epitaxy (OMVPE-16), Keystone, CO, July 21-26, 2013. Y. Tao, A. Yeckel , and J.J. Derby, “Modeling of Particle Engulfment during the Growth of Multicrystalline Silicon for Solar Cells,” Poster presentation, 15th International Summer School on Crystal Growth (ISSCG-15), Gdansk, Poland, August 4-10, 2013. Y. Tao, A. Yeckel , and J.J. Derby, “Modeling of Particle Engulfment during the Growth of Multicrystalline Silicon for Solar Cells,” Poster presentation, 17th International Conference on Crystal Growth and Epitaxy (ICCGE-17), Warsaw, Poland, August 11-16, 2013.
Bibliography Type:	Description: (Last Updated: 09/04/2025)
Abstracts for Journals and Proceedings	Tao Y, Yeckel A, Derby JJ. "Modeling of Particle Engulfment during the Growth of Multicrystalline Silicon for Solar Cells." 19th American Conference on Crystal Growth and Epitaxy and 16th U.S. Biennial Workshop on Organometallic Vapor Phase Epitaxy (OMVPE-16), Keystone, CO, July 21-26, 2013. 19th American Conference on Crystal Growth and Epitaxy and 16th U.S. Biennial Workshop on Organometallic Vapor Phase Epitaxy (OMVPE-16), Keystone, CO, July 21-26, 2013. , Jul-2013
Abstracts for Journals and Proceedings	Tao Y, Yeckel A, Derby JJ. "Modeling of Particle Engulfment during the Growth of Multicrystalline Silicon for Solar Cells." 15th International Summer School on Crystal Growth (ISSCG-15), Gdansk, Poland, August 4-10, 2013. Presentation. 15th International Summer School on Crystal Growth (ISSCG-15), Gdansk, Poland, August 4-10, 2013. , Aug-2013
Abstracts for Journals and Proceedings	Tao Y, Yeckel A, Derby JJ. "Modeling of Particle Engulfment during the Growth of Multicrystalline Silicon for Solar Cells." 17th International Conference on Crystal Growth and Epitaxy (ICCGE-17), Warsaw, Poland, August 11-16, 2013. Presentation. 17th International Conference on Crystal Growth and Epitaxy (ICCGE-17), Warsaw, Poland, August 11-16, 2013. , Aug-2013